



DATA WITH BARAA

WINDOW VALUE FUNCTIONS

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SQL Course | Window Value Functions



$f(x)$ Window Functions

Aggregate

- SUM ()
- AVG ()
- COUNT ()
- MAX ()
- MIN ()

Perform calculations on a set of rows and return a single aggregated value for each row

Rank

- ROW_NUMBER ()
- RANK ()
- DENSE_RANK ()
- NTILE ()
- CUME_DIST ()
- PERCENT_RANK ()

Assign a rank to each row in a window

Value

- LAG ()
- LEAD ()
- FIRST_VALUE ()
- LAST_VALUE ()

Return a **specific value** in a window to be compared with the value of **current row**

Value Functions

Value (Analytics) Functions

LEAD (*expr*,*offset*,*default*)

Returns the value from a previous row

LEAD(Sales,2,0) **OVER** (**ORDER BY** OrderDate)

LAG (*expr*,*offset*,*default*)

Returns the value from a subsequent row

LAG(Sales,2,0) **OVER** (**ORDER BY** OrderDate)

FIRST_VALUE (*expr*)

Returns the first value in a window

FIRST_VALUE(Sales) **OVER** (**ORDER BY** OrderDate)

LAST_VALUE (*expr*)

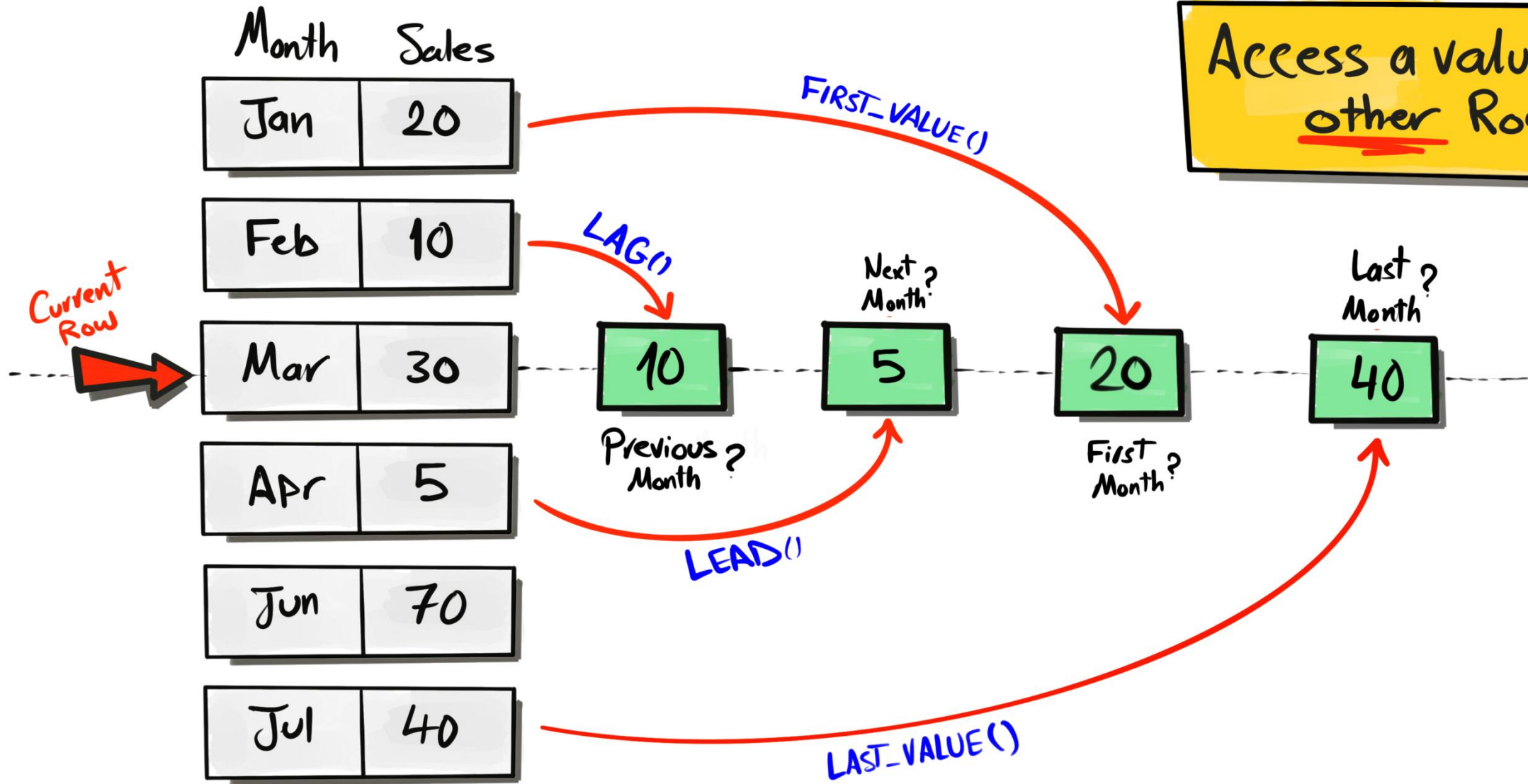
Returns the last value in a window

LAST_VALUE(Sales) **OVER** (**ORDER BY** OrderDate)

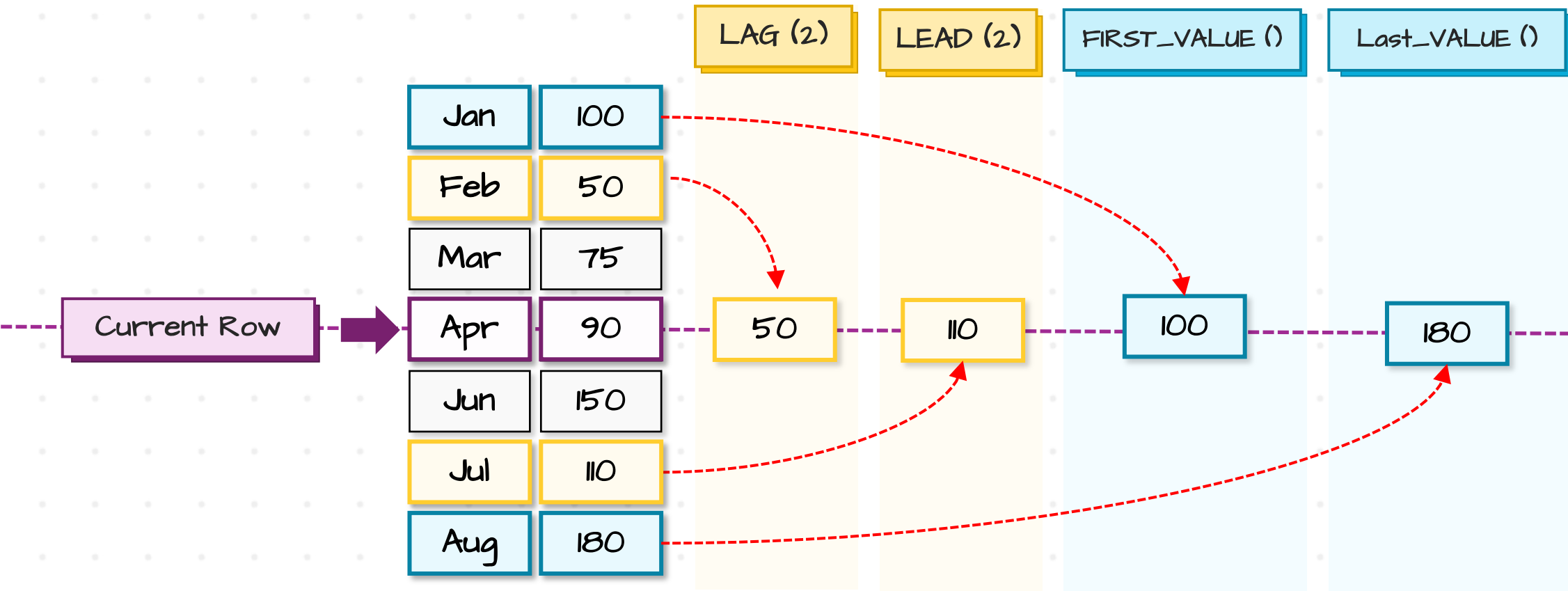
Value Functions

		Expression	Partition Clause	Order Clause	Frame Clause
Value (Analytics) Functions	LEAD (expr,offset,default)	All Data Type	Optional	Required	Not allowed
	LAG (expr,offset,default)				
	FIRST_VALUE (expr)				Optional
	LAST_VALUE (expr)				Should be used

Access a value from
other Row

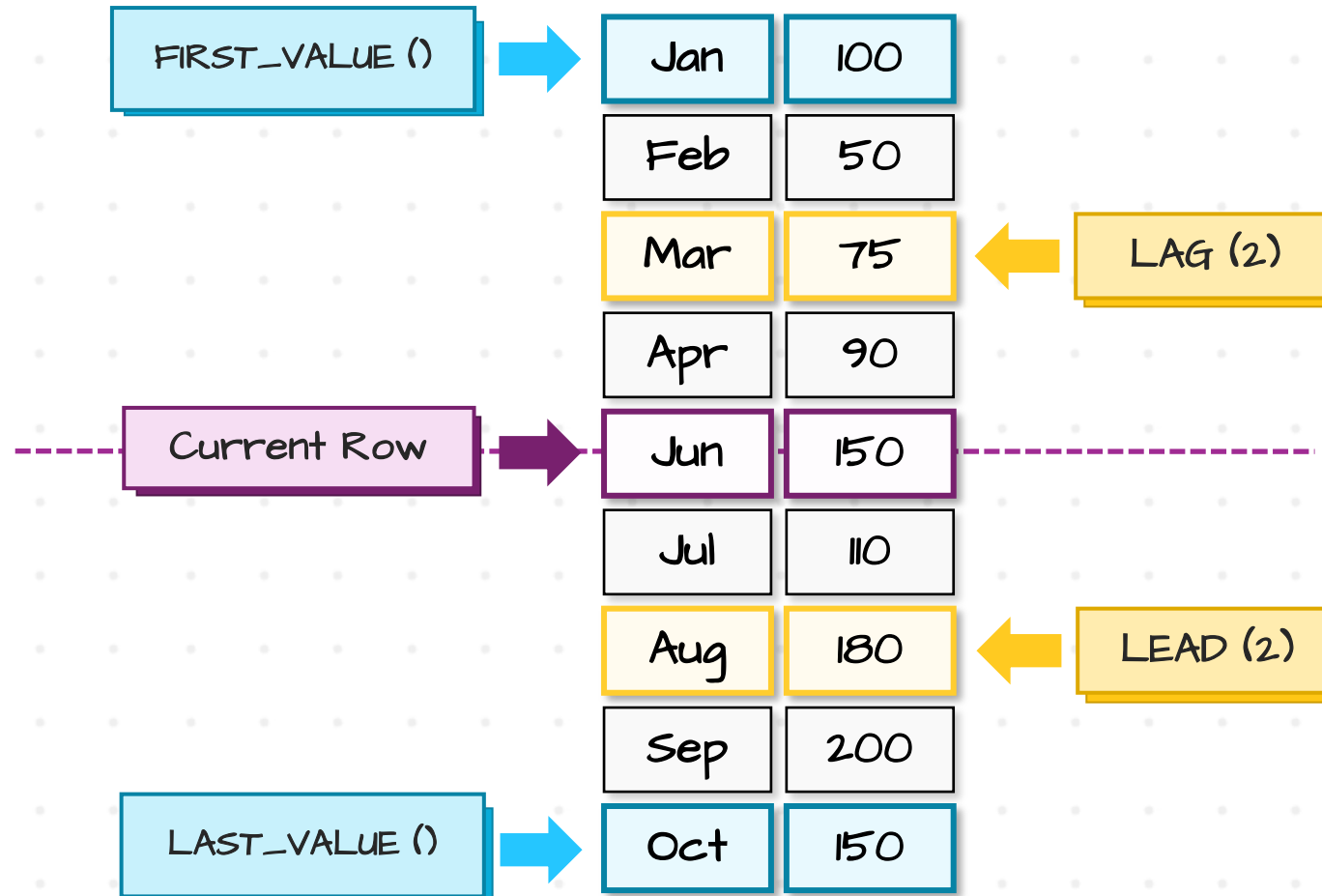


Value Functions



Value Functions

Return a **specific value** in a window to be compared with the value of current row



LEAD & LAG

Partition By
Is **Optional**

Order By
Is **Required**

LEAD (Sales , 2 , 10) **OVER** (PARTITION BY ProductID ORDER BY OrderDate)

Expression
is **required**
(Any Data Type)

Default Value (Optional)

Returns default value if next/previous row is not available!
Default = **NULL**

offset (Optional)

Number of rows forward or backward from current row
default = **1**

LEAD & LAG

`LEAD(Sales) OVER (ORDER BY Month)`

Month	Sales	LEAD
Jan	20	10
Feb	10	30
Mar	30	5
Apr	5	NULL



Find Sales of
the next month

`LAG(Sales) OVER (ORDER BY Month)`

Month	Sales	LAG
Jan	20	NULL
Feb	10	20
Mar	30	10
Apr	5	30



Find Sales of
the previous month

LEAD

Access **Next** Row

```
LEAD (Sales) OVER ( ORDER BY Month)
```

Current Row →	Month	Sales	LEAD
	Jan	20	20
	Feb	10	
	Mar	30	
	Apr	5	

LAG

Access **Previous** Row

```
LAG (Sales) OVER ( ORDER BY Month)
```

First Row has **No** Previous Row !

Current Row →	Month	Sales	LAG
	Jan	20	NULL
	Feb	10	
	Mar	30	
	Apr	5	

LEAD

Access **Next** Row

```
LEAD (Sales) OVER ( ORDER BY Month)
```

Month	Sales	LEAD
Jan	20	20
Feb	10	30
Mar	30	
Apr	5	

Current Row →

LAG

Access **Previous** Row

```
LAG (Sales) OVER ( ORDER BY Month)
```

First Row has **No** Previous Row !

Month	Sales	LAG
Jan	20	NULL
Feb	10	20
Mar	30	
Apr	5	

Current Row →

LEAD

Access **Next** Row

```
LEAD (Sales) OVER ( ORDER BY Month)
```

Month	Sales	LEAD
Jan	20	20
Feb	10	30
Mar	30	5
Apr	5	

Current Row →

LAG

Access **Previous** Row

```
LAG (Sales) OVER ( ORDER BY Month)
```

First Row has **No** Previous Row !

Month	Sales	LAG
Jan	20	NULL
Feb	10	20
Mar	30	10
Apr	5	

Current Row →

LEAD

Access **Next** Row

```
LEAD (Sales) OVER ( ORDER BY Month)
```

Month	Sales	LEAD
Jan	20	20
Feb	10	30
Mar	30	5
Apr	5	NULL

Current Row →

Last Row has **No** Next Row !

LAG

Access **Previous** Row

```
LAG (Sales) OVER ( ORDER BY Month)
```

First Row has **No** Previous Row !

Month	Sales	LAG
Jan	20	NULL
Feb	10	20
Mar	30	10
Apr	5	30

Current Row →

TIME SERIES ANALYSIS

Year-over-Year (YoY)

Analyze the overall growth or decline of the business's performance over time

Month-over-Month (MoM)

Analyze short-term trends and discover patterns in seasonality

FIRST & LAST

`FIRST_VALUE(Sales) OVER (ORDER BY Month)`

	Month	Sales	First
UNBOUNDED PRECEDING	Jan	20	20
	Feb	10	20
	Mar	30	20
Current Row	Apr	5	20

`LAST_VALUE(Sales) OVER (ORDER BY Month)`

	Month	Sales	Last
UNBOUNDED PRECEDING	Jan	20	20
	Feb	10	10
	Mar	30	30
Current Row	Apr	5	5

Default

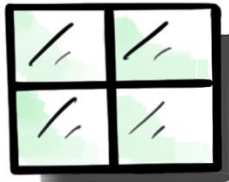
`RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW`

FIRST & LAST

`LAST_VALUE(Sales) OVER (ORDER BY Month
ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING)`

Month	Sales	Last
Jan	20	5
Feb	10	5
Mar	30	5
Apr	5	5

Diagram illustrating the window function `LAST_VALUE(Sales) OVER (ORDER BY Month ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING)`. The table shows the current row (Apr) and the unbounded following rows (Mar, Feb, Jan) used for the calculation. The 'Last' column shows the result of the function, which is 5 for all rows.



SQL WINDOW USE CASES

- Top N Analysis
- Bottom N Analysis
- Identify & Remove Duplicates
- Assign Unique IDs & Pagination
- Data Segmentation
- Data Distribution Analysis
- Equalizing Load Processing
- Overall Analysis
- Total Per Groups Analysis

- Part-to-Whole Analysis
- Time Series Analysis: MoM & YoY
- Time Gaps Analysis: Customer Retention
- Comparison Analysis: Extreme ↗ Highest
↘ Lowest
- Outlier Detection
- Running Total
- Rolling Total
- Moving Average